

Technology Stack

Date	29 June 2026
Team ID	
Project Name	Credit Card Approval Prediction System
Maximum Marks	2 Marks

Define Your Technology Stack:

The Credit Card Approval Prediction System is developed using a machine learning-based architecture with Python and Flask. HTML, CSS, and Bootstrap are used to build a responsive user interface, while machine learning libraries such as Scikit-learn and XGBoost are used for data preprocessing, model training, evaluation, and prediction. IBM Cloud and IBM Watson Machine Learning are used for cloud deployment, enabling secure and scalable real-time credit card approval predictions.

Technology Stack Details

1. Frontend / Client-Side

Technology Chosen : HTML,CSS, Bootstrap

Purpose

Provides a responsive and user-friendly interface for entering applicant details and displaying instant credit card approval predictions.

2. Backend / Server-Side

Technology Chosen : Python, Flask

Justification / Purpose

Processes user requests, loads the trained machine learning model, performs prediction, and returns approval or rejection results quickly.

3. Database / Data Storage

Technology Chosen : CSV Dataset, Pickle (.pkl)

Justification / Purpose

CSV files are used for storing applicant and credit datasets, while the Pickle file stores the trained machine learning model for real-time prediction.

4. Cloud / Hosting / Deployment

Technology Chosen : IBM Cloud, IBM Watson Machine Learning

Justification / Purpose

Provides scalable cloud deployment and enables online access to the prediction system with reliable model hosting.

5. Version Control & CI/CD

Technology Chosen : Git, GitHub

Justification / Purpose

Maintains version history, simplifies code management, and enables collaboration and future project updates.

6. Third-Party APIs / Other Tools

Technology Chosen

- NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn, XGBoost

Justification / Purpose

Supports data preprocessing, visualization, machine learning model training, evaluation, and prediction for the credit card approval system.

Technology Stack Details

S.No	Activity	Technology	Justification / Purpose
1	Frontend / Client-side	HTML, CSS, Bootstrap	Provides a responsive and user-friendly web interface for entering applicant details and displaying prediction results.
2	Backend / Server-side	Python, Flask	Processes user requests, loads the trained machine learning model, and generates approval predictions.
3	Database / Data Storage	CSV Files, Pickle (.pkl)	Stores the training dataset and the trained machine learning model for prediction.
4	Cloud / Hosting / Deployment	IBM Cloud, IBM Watson Machine Learning	Hosts the web application and deploys the trained model for real-time predictions.
5	Version Control & CI/CD	Git, GitHub	Maintains source code versions, enables collaboration, and manages project updates.
6	Third-Party APIs / Other Tools	NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn, XGBoost	Used for data preprocessing, visualization, model training, evaluation, and prediction.